A SEMINORMAL RING IS REDUCED, AND SO IS AN ABSOLUTELY WEAKLY NORMAL RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-seminormal-reduced is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 12101-12105 and proof span 12107-12111 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement's 'A fortiori' clause about absolutely weakly normal schemes is part of the counted statement span and is preserved.

Allowed standard prerequisites (editorial).

- The defining property of seminormality: for a ring A, whenever elements a and b satisfy a squared equals b cubed, there is an element c with c cubed equal to a and c squared equal to b. Used at proof lines 12324-12328, where the property is applied with a nilpotent element and zero. It is the property that a nilpotent would violate.
- A ring is reduced exactly when it has no nonzero element whose square is zero. Used at proof lines 12326-12328. Turns the failure of seminormality for a nilpotent into the reducedness assertion.
- Absolutely weakly normal rings are seminormal. Used at statement lines 12318-12322 for the 'a fortiori' clause. Transfers the reducedness conclusion to absolutely weakly normal rings.

Lemma 0.1. *A seminormal scheme or ring is reduced. A fortiori the same is true for absolutely weakly normal schemes or rings.*

Proof. Let A be a ring. If $a \in A$ is nonzero but $a^2 = 0$, then $a^2 = 0^2$ and $a^3 = 0^3$ and hence A is not seminormal. $\square$

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.